

Accenture Recruiter Interview Prep

Accenture recruiter-style questions and answers tailored to candidate's resume

Candidate: Kishan Sinha | AI/ML & Data Engineer

Resume-Based Recruiter Questions & Answers

Q1: Walk me through your resume, highlighting your key project achievements.

Answer:

I am an AI/ML and Data Engineer with a PG Diploma in Artificial Intelligence (PG-DAI) from CDAC (graduating with 81.47%) and a BE in Electronics and Communication Engineering. My resume is built on three production-grade engineering accomplishments:

1. **QuestionAnswerSystem (RAG chatbot):** I designed a document ingestion system utilizing LangChain (LCEL), Groq API, and FAISS. Key achievements include adding an on-disk embedding cache that reduced document indexing latency from seconds to milliseconds, and implementing post-inference refusal filters to enforce 100% citation integrity.
2. **SentinelGuard (Anomaly Detection):** An unsupervised time-series anomaly detection pipeline using LSTM and GRU Autoencoders to analyze S&P 500 log returns. I built an end-to-end training filter that removed anomalies and high-variance sequences (top 20%) to train the model strictly on "normal" market behavior, and integrated a Jenkins CI/CD process.
3. **Retail Sales Forecasting:** A tabular ML forecasting pipeline comparing Random Forest and XGBoost. I engineered chronological lags, rolling metrics, and calendar indicators, achieving an R^2 score of 0.933. I also configured GitHub Actions for automated deployment.

My stack centers around Python, SQL, PySpark, FastAPI, Docker, and AWS.

Q2: You transitioned from Electronics and Communication Engineering (ECE) to AI/ML and Data Engineering. Why this shift, and how did you prepare?

Answer:

During my BE in Electronics and Communication, I worked heavily with digital signal processing, mathematics, and microcontrollers. I realized that the logic and statistical formulations we used to analyze signals are identical to how machine learning models analyze data. This sparked my interest in software and data intelligence.

To build a strong foundation, I completed a PG-DAI from CDAC. This intensive program gave me formal training in Data Structures & Algorithms, advanced Python, database systems, PySpark for big data processing, and deep learning architectures. I graduated with 81.47%, proving my ability to master complex software and data engineering concepts quickly.

Q3: Accenture works with massive enterprise data. What is your experience with big data engineering using PySpark?

Answer:

I have practical experience building data pipelines using PySpark and Spark SQL to process large datasets. I understand the master-worker execution architecture (Driver, executors, and Cluster Manager) and how lazy evaluation optimizes execution plans (DAGs).

In my projects, I use PySpark to clean nulls, join tables, and calculate rolling aggregates. I am experienced in optimizing Spark operations, specifically utilizing **Broadcast Joins** to bypass heavy network shuffles when joining large tables with smaller lookups, and applying **salting techniques** to resolve data skew. I am also familiar with **Delta Lake** storage layers to bring ACID compliance, schema enforcement, and time-travel querying to data lakes.

Q4: Tell me about your RAG Chatbot project. Why did you choose FAISS and Groq Llama-3.3?

Answer:

The RAG Chatbot is designed to retrieve answers from uploaded PDFs, DOCX, and TXT files.

- **FAISS:** Chosen because it is a highly optimized, light in-memory vector indexing library. It allows us to perform similarity searches in microseconds locally, bypassing the network latency and subscription overhead of cloud-hosted vector databases like Pinecone.
- **Groq Llama-3.3:** Groq's LPU (Language Processing Unit) architecture provides incredibly fast inference speeds (often exceeding 200 tokens per second). This ensures sub-second response latency for the user, which is critical for conversational interfaces.

Q5: How did you implement the custom caching and memory in your RAG system?

Answer:

- **Embedding Cache:** Standard RAG pipelines waste API calls by re-embedding identical files. I wrapped the HuggingFace embeddings in LangChain's **CacheBackedEmbeddings** utilizing a local on-disk **LocalFileStore**. When a user uploads a previously processed document, the system reads cached vectors from disk in milliseconds rather than computing them again.
- **Conversational Memory:** I developed a custom, thread-safe class **InMemoryHistory** inheriting from **BaseChatMessageHistory**. It uses **threading.Lock()** to prevent race conditions in multi-user environments and a bounded **collections.deque** (maxlen=20) to store only the last 10 turns, preventing memory bloat.

Q6: How does SentinelGuard prevent the Autoencoder from learning anomalies, and how did you reduce false positives?

Answer:

- **Training Filter:** Autoencoders detect anomalies by learning to compress and reconstruct normal patterns. If anomalies are present during training, the model reconstructs them easily, failing to flag them at inference. I

built a filter in `lstm_autoencoder.py` that discarded all labeled anomalies and removed the top 20% highest-variance sequences. This trained the model purely on calm baseline data.

- **False Positive Reduction:** Raw reconstruction errors are noisy. I added a three-stage post-processing pipeline:

1. *Noise Floor:* Set the bottom 70% of MSE errors to 0.
2. *Dynamic Threshold:* Calculated the 85th percentile of errors in a rolling window of 60 steps to adapt to volatility shifts.
3. *Temporal Smoothing:* Flagged an anomaly only if at least 3 out of 4 consecutive points exceeded the dynamic threshold, filtering out isolated spikes.

Q7: In your Retail Forecasting project, how did you prevent lookahead bias, and what was your feature engineering strategy?

Answer:

- **Preventing Data Leakage:** Standard random train-test splitting leaks future data into the past. I implemented a chronological time-series split (sorting by date, using the first 80% for training and the newer 20% for testing) to evaluate the model exactly as it would run in production.
- **Feature Engineering:** Standard models can't forecast future dates directly. I engineered:
 - *Calendar markers:* Day of the week, month, and weekend flags to capture cyclical shopping habits.
 - *Lags:* Revenue shifted by 1, 7, 14, and 30 days to capture daily/weekly momentum.
 - *Rolling statistics:* 7-day and 30-day moving averages and standard deviations to smooth noise and capture trends.

Q8: What database technologies have you worked with, and what is your SQL proficiency?

Answer:

I have an Intermediate SQL certification from HackerRank and practical experience working with PostgreSQL and MySQL:

- **Schema Design:** Experienced in normalizing tables up to 3NF to minimize redundancy.
- **Queries:** Conficient in writing complex analytical queries using INNER, LEFT, and FULL joins, subqueries, and aggregation using GROUP BY and HAVING.
- **Advanced Features:** Experienced with SQL Window Functions (`ROW_NUMBER()`, `RANK()`, `DENSE_RANK()`, `SUM() OVER(PARTITION BY...)`) to calculate running totals and find ranks.
- **Optimization:** Understanding how to build Clustered and Non-Clustered indexes to accelerate data retrieval.

Q9: Can you explain how you utilize Docker, Jenkins, and GitHub Actions in your project workflow?

Answer:

DevOps is essential for maintaining and scaling production systems:

- **Docker:** I containerize my backend APIs (FastAPI) and frontends (Streamlit) using multi-stage **Dockerfiles**. This ensures identical environments from development to production.
- **Jenkins:** In *SentinelGuard*, I integrated a Jenkins pipeline that triggers automatically on git push to run unit tests (using **pytest**), verify linting, and compile the code.
- **GitHub Actions:** In *Retail Sales Forecasting*, I configured GitHub Actions CI/CD workflows to automate building, testing, and formatting checks on pull requests.

Q10: How do you build and expose your models? Talk about your API development experience.

Answer:

I use **FastAPI** to build and deploy high-performance REST APIs:

- **Asynchronous Execution:** Utilizing Python's **async** and **await** capabilities to handle concurrent client requests efficiently.
- **Validation:** Exposing Pydantic schemas to automatically validate input JSON payloads, raising structured errors before the request touches database or ML models.
- **Documentation:** Leveraging FastAPI's native integration with OpenAPI to automatically generate interactive Swagger UI endpoints.

Q11: What is your experience with AWS, and how do you deploy applications to the cloud?

Answer:

I utilize AWS to host and run data pipelines and applications:

- **EC2:** Hosting dockerized web containers (FastAPI/Streamlit) on virtual servers.
- **S3:** Object storage for datasets, raw text files, and saved ML model pickles.
- **IAM:** Structuring secure access roles and JSON policies to grant specific permissions (e.g., allowing an EC2 instance to read/write to a designated S3 bucket without exposing credentials).
- **VPC:** Configuring security groups and subnets to isolate database engines in private subnets while exposing web interfaces in public subnets.

Q12: How would you transition your AWS skills if an Accenture client project required you to work on Microsoft Azure or Google Cloud?

Answer:

I have a solid cloud engineering mindset, which makes transitioning straightforward. Most cloud providers offer equivalent services. For example:

- AWS EC2 maps directly to Azure Virtual Machines and GCP Compute Engine.
- AWS S3 maps directly to Azure ADLS Gen2 (Blob Storage) and GCP Cloud Storage.
- AWS Lambda maps directly to Azure Functions and GCP Cloud Functions.

Since I bundle my applications using **Docker**, the underlying application remains independent of the cloud provider. I would quickly map my understanding of IAM roles and VPC configurations to Azure Active Directory

and Virtual Networks, allowing me to start contributing in a few days.

Q13: Accenture values collaborative and continuous learning. How do you keep up with new technologies?

Answer:

I am a self-driven learner. My transition from an ECE degree to graduating with 81% in CDAC's AI program is a testament to my ability to learn complex technologies quickly.

Currently, I follow research publications, read engineering blogs (like HuggingFace, OpenAI, and LangChain), and build projects to test new frameworks. For instance, I am currently exploring agentic workflows using LangGraph and testing advanced vector retrieval methods (such as Hybrid Search and Re-ranking) to keep my skills sharp.

Q14: How do you resolve conflicts within a software engineering team?

Answer:

I resolve conflicts through **data-driven discussion** and **collaborative alignment**:

1. *Listen First*: I ensure both parties explain their viewpoints fully without interruption.
2. *Review the Data*: If we are debating which model or database structure to use, I suggest writing a quick benchmark script (e.g., comparing database latency or model inference speeds) to let the numbers guide the choice.
3. *Consensus*: I align with the team's objective, ensuring we agree on the best decision for the client, and document the decision to keep development moving.

Q15: What is your work authorization status, and are you willing to relocate?

Answer:

I am an Indian citizen based in India, with full work authorization to take on roles immediately. Yes, I am completely willing to relocate to any of Accenture's key offices or client sites across India. I am highly flexible and excited to join your collaborative teams.